

Shou-Tzu Han (Debra Han)

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Research Profile

Computer Science Ph.D. student and Graduate Research Assistant at Wayne State University researching **language-model reliability and improvement**. Experienced in deep learning and reproducible GPU/HPC experimentation using **PyTorch** and **Hugging Face Transformers**, with work spanning mechanistic analysis, model evaluation, and targeted repair of reasoning failures.

Research Interests

- **LLM Agent Safety:** Locating and attributing failures in LLM agent reasoning during multi-step, safety-critical tasks, and compiling verified failures into runtime checks.
- **LLM Robustness and Reasoning:** Evaluating reasoning stability under meaning-preserving perturbations and diagnosing performance–confidence mismatches.
- **Mechanistic Interpretability:** Using activation analysis, patching, and ablation to identify internal components associated with model failures.
- **Agentic Retrieval and Scientific Discovery:** Building multi-step retrieval systems for literature comparison, contribution analysis, and research-gap identification.

Publications

- **Shou-Tzu Han**, Rodrigue Rizk, and KC Santosh. *Fragile Reasoning: A Mechanistic Analysis of LLM Sensitivity to Meaning-Preserving Perturbations*. Preprint. arXiv:2604.01639
- **Shou-Tzu Han**. *Novelty-Aware Agentic Retrieval: Comparing Research Contributions Through Structured Multi-Step Reasoning*. Preprint. arXiv:2606.22151
- **Shou-Tzu Han**. *Narrative-Centered Emotional Reflection: An Early Prototype for AI-Supported Emotional Self-Reflection*. Preprint. arXiv:2504.20342

Research Experience

Graduate Research Assistant

Wayne State University, Trustworthy AI Lab
Advisors: Dongxiao Zhu and Sooin Kim

August 2026–Present

LLM Agent Safety Research

- Research where and why LLM agent reasoning fails during multi-step, safety-critical tasks, spanning action selection, tool use, and decision-making under injected or adversarial context.
- Develop an outcome-grounded auditing method that commits checkable predictions before an action, then uses the real execution result to localize the failing reasoning step and compile it into a runtime check.
- Design fault-injection and component-removal experiments on agent benchmarks to measure whether the method attributes failures to the correct cause.

AI-Supported Mentoring Systems

- Develop AI-supported mentoring systems aimed at improving retention and educational experiences for veteran engineering students.
- Build LLM-based mentoring components that deliver structured, context-aware guidance and relevant academic resources.
- Contribute to system prototyping, experimental design, data analysis, and evaluation of AI-generated support.

Graduate Research Assistant

University of South Dakota, AI Research Lab
Advisors: KC Santosh and Rodrigue Rizk

January 2025–May 2025

LLM Robustness and Mechanistic Interpretability

- Conducted GPU/HPC experiments using PyTorch and Hugging Face Transformers to evaluate LLM reasoning stability under meaning-preserving perturbations.
- Applied activation analysis, patching, and ablation to diagnose reasoning failures; this work resulted in the *Fragile Reasoning* preprint (arXiv:2604.01639).

Research Projects

Fragile Reasoning: A Mechanistic Analysis of LLM Sensitivity to Meaning-Preserving Perturbations

- Evaluated Mistral-7B, Llama-3-8B, and Qwen2.5-7B on 677 paired GSM8K problems, revealing answer-flip rates of 28.8%–45.1% under meaning-preserving perturbations.
- Developed the **Mechanistic Perturbation Diagnostics (MPD)** framework, integrating logit-lens analysis, activation patching, component ablation, and the novel **Cascading Amplification Index (CAI)**.
- Identified localized, distributed, and entangled failure modes, demonstrating substantial architectural differences in failure localization and recoverability.
- Evaluated steering vectors and layer fine-tuning as targeted repair methods using **Python, PyTorch, Hugging Face Transformers, GPUs, and SLURM**.

Novelty-Aware Agentic Retrieval: Comparing Research Contributions Through Structured Multi-Step Reasoning

- Built an agentic retrieval system over a 100-paper corpus using six components: query analysis, iterative retrieval, ranking, contribution extraction, comparison, and answer generation.
- Designed structured contribution records and a three-pass comparison agent to identify paper-level overlaps, methodological differences, and research gaps.
- Developed a **problem–method gap matrix** that generates citation-grounded evidence for unexplored combinations within a research corpus.
- Achieved a mean Precision@5 of 0.980, nDCG@5 of 0.739, and 84.0% schema compliance across ten evaluation queries.
- Implemented the system using **Python, GPT-4o, FAISS/Chroma, SentenceTransformers**, and structured prompting.

Technical Skills

- **Machine Learning & LLMs:** PyTorch, TensorFlow, Hugging Face Transformers, OpenAI API, prompt design, model evaluation, reasoning analysis
- **Interpretability & Reliability:** Logit lens, activation patching, component ablation, steering vectors, perturbation analysis, failure diagnosis
- **Agentic AI & Retrieval:** RAG, FAISS, Chroma, BM25, SentenceTransformers, ReAct-style agents, structured prompting
- **Programming & Data:** Python, Java, JavaScript, TypeScript, SQL, Bash, pandas, NumPy, Matplotlib
- **Computing & Development:** SLURM, HPC, CUDA/GPU computing, Linux, Git, Docker, FastAPI, Flask, React, Vercel

Education

Wayne State University, Detroit, Michigan
Ph.D. in Computer Science
Graduate Research Assistant, Trustworthy AI Lab

August 2026–Present

Boston University, Boston, Massachusetts
Master of Science in Computer Science

September 2021–August 2023

Soochow University, Taipei, Taiwan
Bachelor of Arts in Political Science

September 2015–June 2020